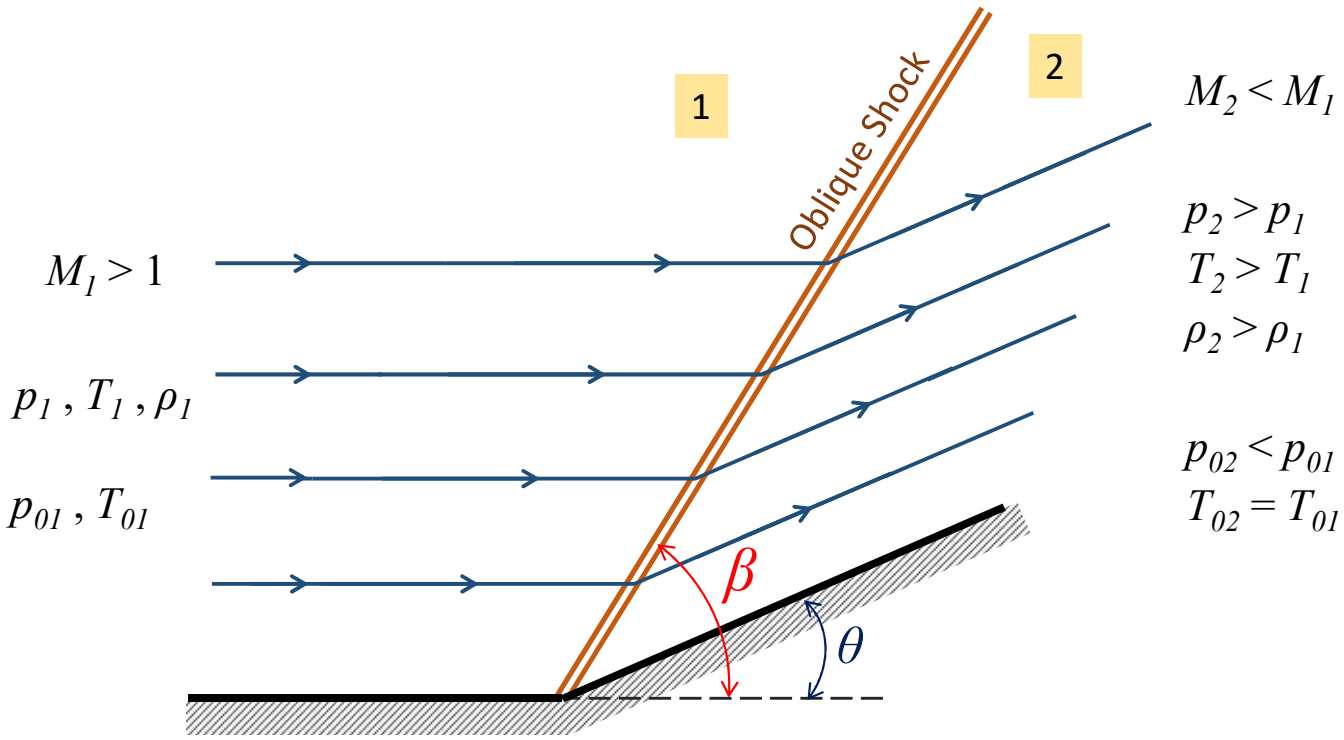
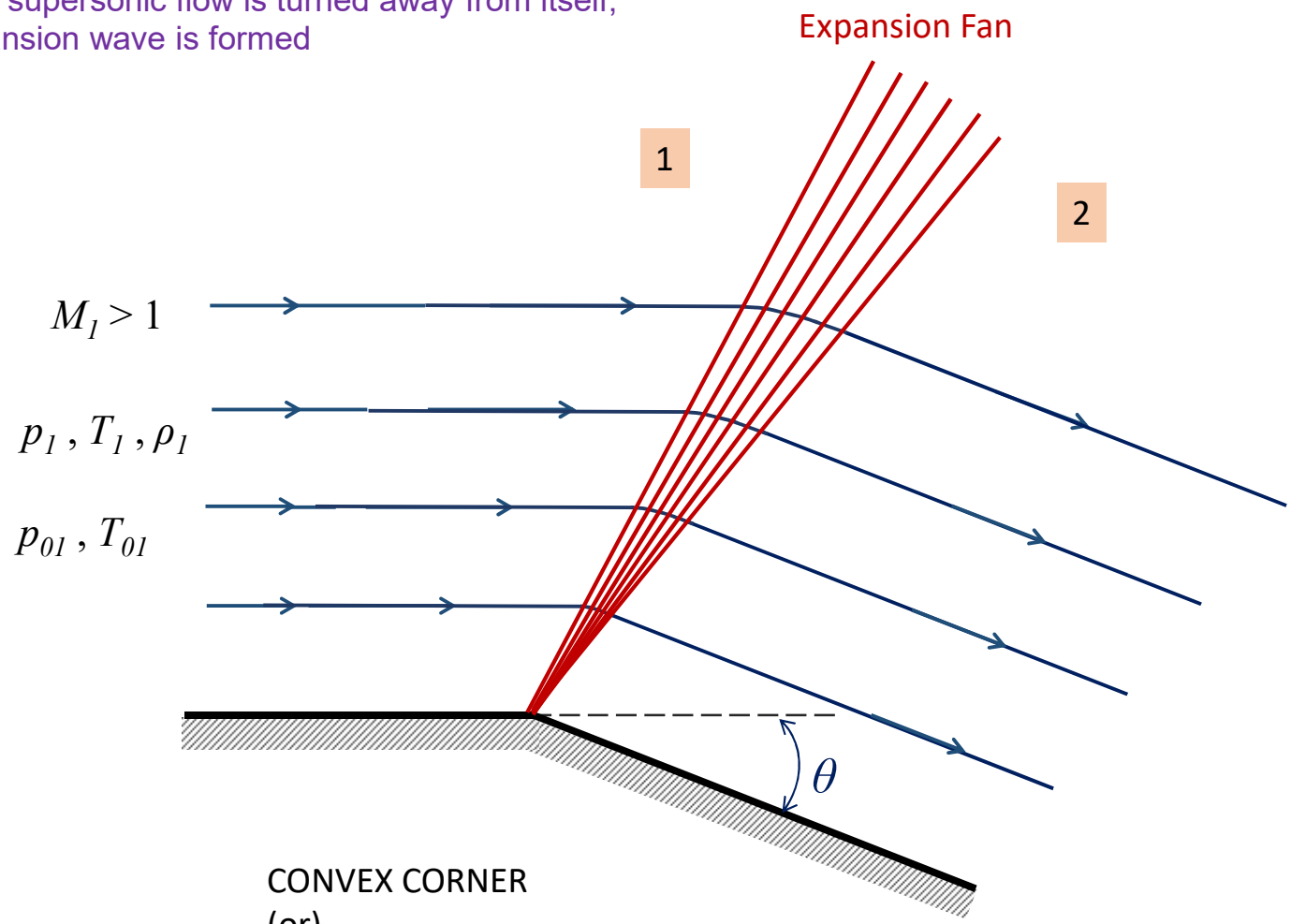
PRANDTL-MEYER EXPANSION WAVES

Quick Recap: When a supersonic flow is turned into itself, an oblique shock is formed



CONCAVE CORNER
(or)
Compression Corner

When a supersonic flow is turned away from itself, an expansion wave is formed



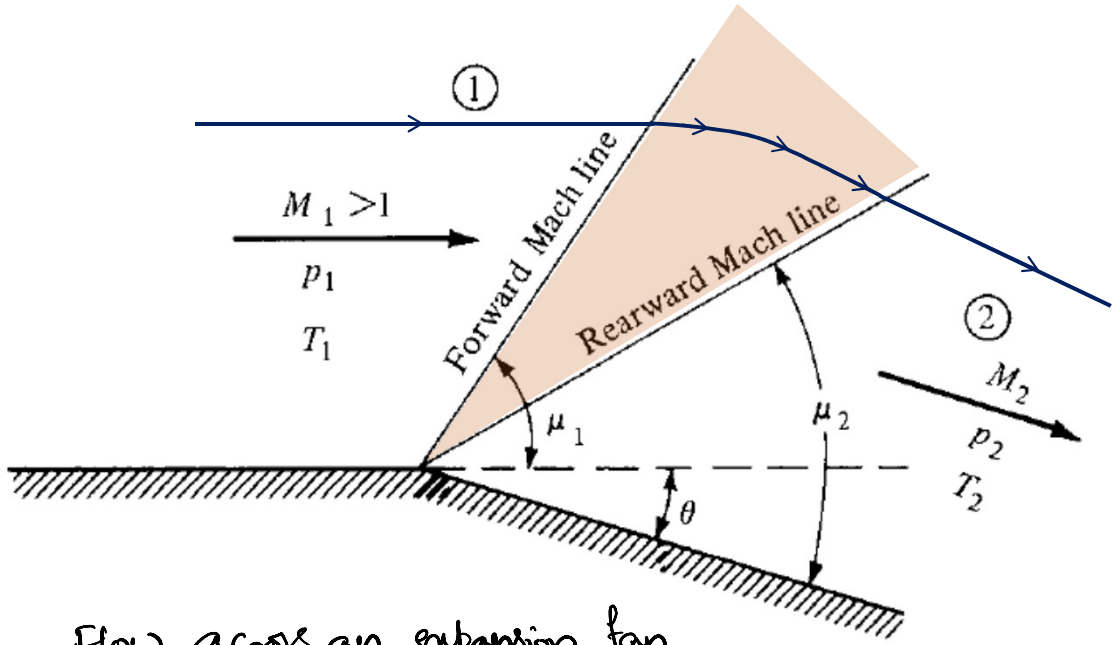
CONVEX CORNER
(or)
Expansion Corner

$M_2 > M_1$

$p_2 < p_1$
$T_2 < T_1$
$\rho_2 < \rho_1$

$p_{02} = p_{01}$
$T_{02} = T_{01}$

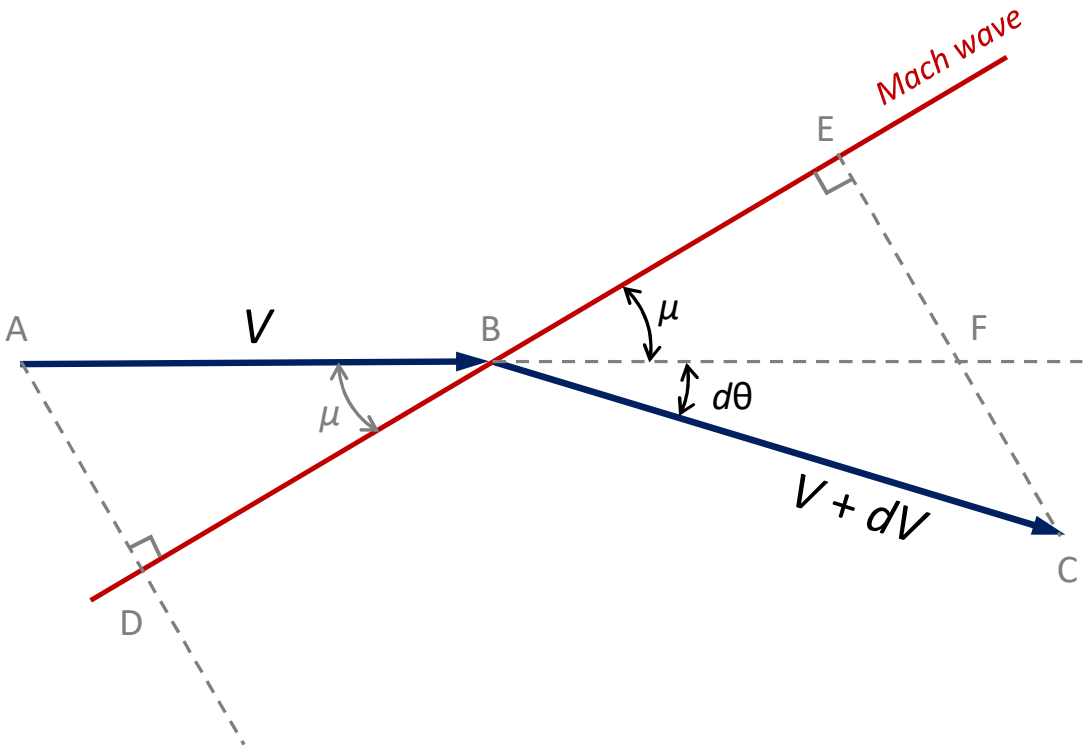
- The expansion fan is a *continuous* expansion region that can be visualized as an infinite number of Mach waves, each making the Mach angle μ with the local flow direction.
- The expansion fan is bounded upstream by a Mach wave which makes the angle μ_1 w.r.t. the upstream flow, where $\mu_1 = \sin^{-1}(1/M_1)$.
- The expansion fan is bounded downstream by another Mach wave which makes the angle μ_2 w.r.t. the downstream flow, where $\mu_2 = \sin^{-1}(1/M_2)$.
- Since the expansion through the wave takes place across a continuous succession of Mach waves, and since $ds = 0$ for each Mach wave, the expansion is *isentropic*.



⇒ Flow across an expansion fan is Isentropic ⇒ Stagnation Conditions are constant across an expansion fan.

Expansion wave Solution: Given the upstream flow (region 1) and the deflection angle θ, calculate the downstream flow (region 2)

- Consider a Mach wave produced by an infinitesimally small flow deflection $d\theta$ ($d\theta \to 0$)
- Mach wave forms at an angle μ to the upstream flow
- Velocity changes from V to V+dV across the Mach wave



From similar triangles ADB and BEF ,

BD = BE

$$V\cos\mu = (V + dV)\cos(\mu + d\theta)$$

$$\frac{V + dV}{V} = \frac{\cos\mu}{\cos(\mu + d\theta)}$$

$$1 + \frac{dV}{V} = \frac{\cos\mu}{\cos\mu\cos d\theta - \sin\mu\sin d\theta}$$

For small dθ, we can make the small-angle assumptions sin dθ ≈ dθ and cos dθ ≈ 1.

$$1 + \frac{dV}{V} = \frac{\cos\mu}{\cos\mu - d\theta \sin\mu} = \frac{1}{1 - d\theta \tan\mu}$$

Note that the function $1/(1-x)$ can be expanded in a power series (for $x < 1$) as

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$$

$$1 + \frac{dV}{V} = 1 + d\theta \tan\mu \cdot$$ (*ignoring terms of second order and higher*)

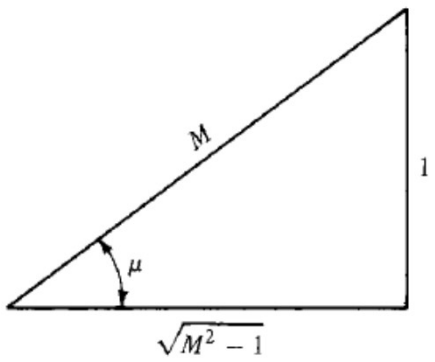
$$d\theta = \frac{dV/V}{\tan\mu}$$

$$\boxed{d\theta = \sqrt{M^2 - 1}\,\frac{dV}{V}}$$ ------ eqn (1)

differential equation for the flow inside the expansion wave

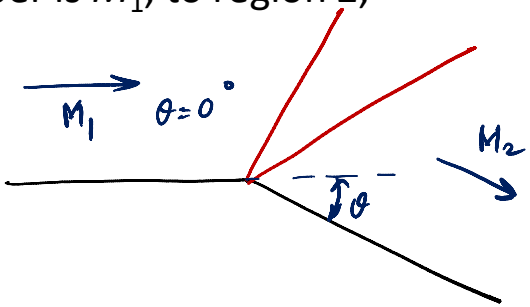
$\mu = \sin^{-1}\left(\frac{1}{M}\right)$

$\Rightarrow \sin\mu = \frac{1}{M}$



$$\tan\mu = \frac{1}{\sqrt{M^2 - 1}}$$

Integrating from region 1, where the deflection angle is zero and the Mach number is M_1, to region 2, where the deflection angle is θ and the Mach number is M_2.



$$\int_0^{\theta} d\theta = \theta = \int_{M_1}^{M_2} \sqrt{M^2 - 1}\,\frac{dV}{V}$$ ------ eqn (2)

From the definition of Mach number, $M = V/a$, we have $V = Ma$, or

$$\ln V = \ln M + \ln a$$

$$\frac{dV}{V} = \frac{dM}{M} + \frac{da}{a}$$ ------ eqn (3)

From isentropic relations, $$\left(\frac{a_0}{a}\right)^2 = \frac{T_0}{T} = 1 + \frac{\gamma - 1}{2} M^2$$

$$a = a_0 \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-1/2}$$

$$\frac{da}{a} = -\left(\frac{\gamma - 1}{2}\right) M \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-1} dM \qquad \text{------ eqn (4)}$$

eqn (4) in eqn (3) gives $$\frac{dV}{V} = \frac{1}{1 + [(\gamma - 1)/2]M^2} \frac{dM}{M} \qquad \text{------ eqn (5)}$$

eqn (5) in eqn (2) gives

$$\theta = \int_{M_1}^{M_2} \frac{\sqrt{M^2-1}}{1+[(\gamma-1)/2]M^2}\frac{dM}{M} = \left[\sqrt{\frac{\gamma+1}{\gamma-1}}\tan^{-1}\sqrt{\frac{\gamma-1}{\gamma+1}(M^2-1)} - \tan^{-1}\sqrt{M^2-1}\right]_{M_1}^{M_2}$$

$$\boxed{\theta = \nu(M_2) - \nu(M_1)}$$

where,

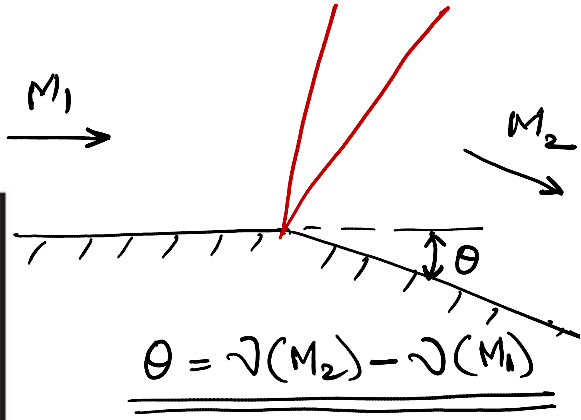
$$\boxed{\nu(M) = \sqrt{\frac{\gamma+1}{\gamma-1}}\tan^{-1}\sqrt{\frac{\gamma-1}{\gamma+1}(M^2-1)} - \tan^{-1}\sqrt{M^2-1}}$$

ν(M) is called the Prandtl-Meyer function

TABLE 9.1

Prandtl–Meyer functions and Mach angles

M (Mach number)	α (Mach angle μ)	$\omega(M)$ (P-M function $\nu(M)$)		
		$\gamma = 1.3$	$\gamma = 1.4$	$\gamma = 1.67$
1.62	38.12	16.556	15.452	13.131
1.64	37.57	17.203	16.043	13.611
1.66	37.04	17.850	16.633	14.089
1.68	36.53	18.500	17.222	14.564
1.70	36.03	19.144	17.810	15.037



How to find the properties in region 2 from the known properties in region 1 and the known deflection angle θ?

1. For the given M_1, obtain $\nu(M_1)$ from Prandtl-Meyer tables

2. Calculate $\nu(M_2)$ *from the relation*

$$\nu(M_2) = \nu(M_1) + \theta$$

3. Obtain M_2 from Prandtl-Meyer tables corresponding to the value of $\nu(M_2)$

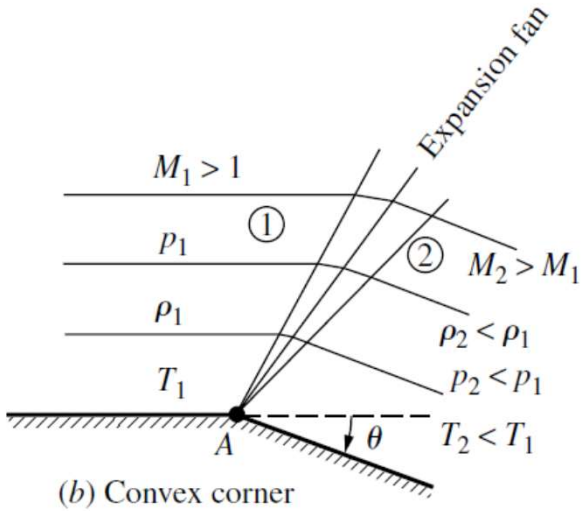
4. The expansion wave is isentropic; hence, p_0 and T_0 are constant through the wave. That is, $T_{02} = T_{01}$ and $p_{02} = p_{01}$.

5. From Isentropic tables, obtain p_1/p_{01} and T_1/T_{01} corresponding to M_1. Similarly, obtain p_2/p_{02} and T_2/T_{02} corresponding to M_2

6. The ratios p_2/p_1 and T_2/T_1 can be obtained as

$$\frac{p_2}{p_1} = \frac{p_2 / p_{02}}{p_1 / p_{01}}$$

$$\frac{T_2}{T_1} = \frac{T_2 / T_{02}}{T_1 / T_{01}}$$



(b) Convex corner